UNIT 3

FORWARDS AND FUTURES

1. INTRODUCTION TO FUTURES

1.1 Meaning of futures

A futures contract is a standardised agreement traded on an organised exchange to buy or sell an underlying asset at an agreed price for settlement on a specified future date, subject to the contract's terms and exchange rules.

In simple language, a futures contract allows a person to take a position today based on the price of an asset at a future date.

The underlying asset may be:

- Shares of companies
- Stock market indices
- Foreign currencies
- Government bonds
- Gold and silver
- Crude oil and other commodities
- Other eligible financial instruments

Futures contracts are important because prices in financial markets change continuously. These changes create uncertainty for investors, companies, importers, exporters and financial institutions.

Futures help market participants manage some of this uncertainty.

1.2 Simple illustration of a futures contract

Suppose a jewellery manufacturer expects to purchase gold after one month.

The current market price of gold is ₹7,000 per gram. The manufacturer is worried that the price may rise in the coming month.

For illustration, assume the manufacturer takes a futures position at ₹7,100 per gram.

Two possible situations may occur.

Situation 1: Gold prices increase

After one month, the relevant futures price rises to ₹7,400 per gram.

The long futures position gains ₹300 per gram, before costs and other settlement adjustments.

Situation 2: Gold prices decrease

After one month, the relevant futures price falls to ₹6,800 per gram.

The long futures position loses ₹300 per gram.

This example illustrates how a futures position responds to changes in price. Actual gold futures have specified contract sizes, expiry dates and settlement rules, so this is a simplified illustration rather than a description of a particular exchange contract.

The manufacturer may use the futures position to offset some of the risk associated with rising gold prices. This is known as hedging.

1.3 Why are futures contracts needed?

Consider the following practical situations.

Situation	Problem faced	Possible use of futures
An investor owns shares	Share prices may fall	Sell suitable stock index futures to reduce market exposure
An importer must pay US dollars	The dollar may become more expensive	Use an appropriate currency derivative to manage exchange-rate risk
An airline expects to purchase fuel	Fuel prices may increase	Use suitable fuel-related futures where available
A fund manager manages a large portfolio	The market outlook is uncertain	Adjust market exposure using index futures
A trader expects prices to rise	A price increase may create a trading opportunity	Buy futures, accepting the associated risk

Futures contracts are therefore used for both risk management and trading.

1.4 Objectives of futures contracts

1. Hedging

Hedging means taking a position to reduce the risk of an adverse price movement.

For example, an investor who owns shares may sell stock index futures to reduce the impact of a broad market decline.

2. Speculation

Speculation means taking a financial position based on an expectation that prices will rise or fall.

For example, a trader who expects an index to rise may buy index futures and attempt to earn a profit.

Speculation involves risk because the market may move in the opposite direction.

3. Arbitrage

Arbitrage involves attempting to earn a profit from price differences between related instruments or markets.

For example, a trader may compare the prices of an index futures contract and the corresponding underlying share portfolio.

However, the price difference must be large enough to justify financing costs, transaction costs and other risks.

4. Price discovery

Futures prices provide information about market expectations and current supply-and-demand conditions.

Investors, companies and analysts use this information while making financial decisions.

5. Portfolio management

Portfolio managers can use futures to change their market exposure without buying or selling every individual share in a portfolio.

1.5 Important terms used in futures markets

Term	Meaning in simple language
Underlying asset	The asset or index on which the futures contract is based
Futures price	The price at which the futures contract is traded
Spot price	The current market price of the underlying asset
Expiry date	The date on which the contract expires
Lot size	The quantity represented by one contract
Long position	Buying a futures contract
Short position	Selling a futures contract
Margin	Funds or collateral required to support a futures position
Mark-to-market	Calculating and settling gains or losses as prices change
Open interest	The number of outstanding contracts that remain open
Basis	Spot price minus futures price
Clearing house	An organisation that manages clearing, margins and settlement

2. FORWARD AND FUTURES CONTRACTS

2.1 Meaning of a forward contract

A forward contract is a private agreement between two parties to buy or sell an asset at a price agreed today for a future date.

Forward contracts are commonly arranged directly between businesses, banks and other financial institutions. They are generally traded in the over-the-counter (OTC) market rather than on an organised exchange.

The parties can customise the contract according to their requirements.

Example: Forward contract used by an importer

- An Indian importer must pay US$100,000 to a foreign supplier after three months.
- The importer is concerned that the US dollar may appreciate against the Indian rupee.
- Suppose the importer and a bank agree on a forward exchange rate of ₹85 per US dollar.

The planned rupee payment is:

$$\$100{,}000 \times ₹85 = ₹85{,}00{,}000$$

- The importer can plan for a payment of ₹85 lakh under the agreed exchange-rate terms, subject to the contract conditions.
- If the market exchange rate later rises to ₹88 per dollar, the forward agreement may protect the importer from the increase in the rupee cost.
- If the market exchange rate falls to ₹82 per dollar, the importer may not benefit from the lower market rate because the forward agreement remains binding under its terms.

This example shows that a forward contract provides certainty but may also mean giving up the benefit of favourable market movements.

2.2 Meaning of a futures contract

A futures contract is similar in purpose to a forward contract, but it is standardised and traded on an organised exchange.

The exchange specifies the contract terms, including the underlying asset, contract quantity, expiry and settlement procedure.

Margins and clearing-house arrangements help manage the risk that a participant may fail to meet their obligations.

Example

Suppose a trader expects the Nifty 50 index to rise.

The trader buys a Nifty 50 futures contract at an assumed price of 25,000 points.

If the futures price rises to 25,200 points, the trader gains 200 index points multiplied by the applicable contract multiplier.

If the futures price falls to 24,800 points, the trader loses 200 index points multiplied by the contract multiplier.

The actual monetary gain or loss depends on the contract specifications.

2.3 Differences between forward and futures contracts

Basis	**Forward contract**	**Futures contract**
Meaning	Private agreement for a future transaction	Standardised exchange-traded agreement
Trading place	OTC	Organised exchange
Contract size	Customised	Standardised
Contract date	Flexible, as agreed	Selected from exchange-listed expiries
Counterparty risk	Depends on the parties and collateral arrangements	Managed through clearing and margin arrangements
Settlement	Often at maturity, depending on the agreement	Typically involves daily mark-to-market and final settlement
Liquidity	May be difficult to transfer or exit	Often easier to close in actively traded contracts
Transparency	Contract terms may be private	Trading prices and contract terms are generally more transparent
Flexibility	High	Lower because terms are standardised
Typical use	Specific business exposures	Hedging, trading and portfolio management

Easy way to remember

- A forward contract is like a private arrangement between two parties.
- A futures contract is like a standardised agreement traded through an organised exchange.
- Both can be used to manage financial risk, but their trading and settlement arrangements differ.

2.4 Advantages of forward contracts

1. Contract terms can be customised.
2. The amount and maturity can match a specific business requirement.
3. They can be useful for managing known foreign-currency payments and receipts.
4. They can provide certainty about future transaction prices or exchange rates.

2.5 Limitations of forward contracts

1. There may be counterparty risk.
2. Exiting the agreement early may be difficult or costly.
3. There may be less transparency in pricing.
4. The parties may be exposed to credit and liquidity risks.

2.6 Advantages of futures contracts

1. Standardised contracts make trading easier.
2. Exchange trading generally improves price transparency.
3. Many contracts provide opportunities to enter and exit positions before expiry.
4. Clearing and margin systems help manage counterparty risk.
5. Futures can be used for hedging and portfolio management.

2.7 Limitations of futures contracts

1. Standardised contract sizes may not exactly match a business's exposure.
2. Margin calls may require additional funds.
3. Leverage can magnify losses.
4. Futures may not perfectly offset the underlying exposure.
5. Brokerage, taxes and other charges reduce returns.

3. FUTURES MARKETS

3.1 Meaning of futures markets

A futures market is a marketplace in which standardised futures contracts are bought and sold.

These markets allow participants to manage price risk, express market expectations and trade

3.2 Major participants in futures markets

1. Hedgers

Hedgers participate in futures markets to reduce an existing or expected financial risk.

Example: An investor holding a share portfolio sells suitable index futures to protect against a potential market decline.

The objective is primarily risk reduction rather than making a speculative profit.

2. Speculators

Speculators accept market risk in the hope of earning a profit.

Example: A trader expects the Nifty 50 index to rise and buys index futures.

If prices increase, the trader may earn a profit. If prices decline, the trader may suffer a loss.

3. Arbitrageurs

Arbitrageurs look for price differences between related financial instruments.

Example: A trader compares an index futures price with the theoretical futures price calculated using the spot index, interest rates, expected dividends and time to expiry.

If a sufficiently attractive pricing difference exists, the trader may establish related positions.

Arbitrage is not guaranteed to be profitable because costs, execution problems and market risks can eliminate the apparent opportunity.

4. Brokers and trading members

Brokers provide access to the exchange trading system and execute orders for clients, subject to applicable rules.

5. Clearing corporations

Clearing corporations manage clearing and settlement, collect margins and apply risk-management procedures.

6. Regulators

Regulators establish and supervise the rules intended to promote fair, orderly and transparent markets.

3.3 Functions of futures markets

1. Risk management: Businesses and investors can reduce exposure to unfavourable price movements.
2. Price discovery: Futures prices reflect market trading and expectations.
3. Liquidity: Active contracts may allow participants to enter and exit positions more easily.
4. Efficient portfolio adjustment: Investors can change market exposure without immediately trading every underlying asset.
5. Support for arbitrage: Related futures and cash markets can be compared, helping maintain pricing relationships.

3.4 Why are futures markets important to the economy?

Futures markets can help businesses plan future costs, enable investors to manage portfolio risk and provide information about prices.

For example, an exporter who manages foreign-exchange exposure may be better able to estimate future cash flows. An investor who manages equity exposure through index futures may be able to reduce the impact of a temporary market decline.

However, futures markets do not eliminate economic uncertainty. They provide tools to transfer, manage or assume risk.

4. MECHANICS OF FUTURES MARKETS

The following steps describe a typical exchange-traded futures transaction.

Step 1: Selecting a contract

The trader selects the underlying asset and expiry month.

For example, the trader may select a stock index futures contract expiring in a specified month.

Step 2: Placing an order

The trader places a buy or sell order through a broker or trading member.

A buy order opens a long position if the trader does not already have an offsetting short position. A sell order can open a short position or reduce an existing long position.

Step 3: Matching the order

The exchange's trading system matches compatible buy and sell orders.

Step 4: Providing margin

The trader must maintain the required margin according to the applicable exchange and clearing rules.

Margin is not the full purchase price of the underlying asset. It acts as a financial safeguard for the obligations arising from the position.

Step 5: Daily valuation and settlement

The futures position is valued using the relevant settlement price. Gains and losses are processed through the mark-to-market system.

Step 6: Closing or settling the contract

The trader may close the position before expiry by taking an equal and opposite position in the same contract.

Alternatively, the position continues until expiry and follows the specified settlement procedure.

Profit and loss on a long futures position

A long futures position is opened by buying a futures contract.

The simplified profit or loss is:

$$\text{P/L} = (F_1 - F_0) \times Q$$

Example

A trader buys a futures contract at ₹200 for 50 units.

Later, the trader closes the position at ₹218.

$$\text{Profit} = (218 - 200) \times 50$$

$$= 18 \times 50$$

$$\boxed{\text{Profit} = ₹900}$$

If the closing price were ₹188 instead:

$$\text{Loss} = (188 - 200) \times 50$$

$$= -12 \times 50$$

$$\boxed{\text{Loss} = ₹600}$$

Profit and loss on a short futures position

A short futures position is opened by selling a futures contract.

The simplified profit or loss is:

$$\text{P/L} = (F_0 - F_1) \times Q$$

Example

A trader sells a futures contract at ₹200 for 50 units.

Later, the trader buys it back at ₹188.

$$\text{Profit} = (200 - 188) \times 50$$

$$= 12 \times 50$$

$$\boxed{\text{Profit} = ₹600}$$

If the closing price rises to ₹218:

$$\text{Loss} = (200 - 218) \times 50$$
$$= -18 \times 50$$
$$\boxed{\text{Loss} = ₹900}$$

Important observation

- A long position generally benefits when the futures price rises.
- A short position generally benefits when the futures price falls.

The trader must consider the possibility of losses in either position.

Comparison between long and short positions

Feature	Long position	Short position
Initial transaction	Buy futures	Sell futures
Market expectation	Generally bullish	Generally bearish
Benefits when	Futures prices rise	Futures prices fall
Loses when	Futures prices fall	Futures prices rise
Possible use	Managing risk of future price increases	Managing risk of falling asset values
Example	Trader buys index futures expecting a rise	Investor sells index futures to hedge a portfolio

Why do traders close futures positions?

Traders may close their positions for several reasons:

1. They have achieved their expected profit.
2. They want to prevent further losses.
3. Their market expectations have changed.
4. They no longer need the hedge.
5. The contract is approaching expiry.
6. They want to transfer their exposure to a later-expiry contract.

HEDGING STRATEGIES USING FUTURES

Meaning of hedging

Hedging is a strategy used to reduce the risk of an unfavourable price movement.

It involves taking a financial position that is expected to offset some of the gains or losses on an existing asset, liability or expected transaction.

A simple everyday analogy is carrying an umbrella when rain is expected. The umbrella does not stop the rain, but it can reduce the inconvenience caused by it.

Similarly, futures cannot prevent market prices from changing, but a properly designed futures

Why do businesses and investors hedge?

Consider the following examples:

- An investor fears a fall in the stock market.
- A manufacturer expects the cost of a commodity to rise.
- An exporter expects to receive foreign currency.
- A fund manager wants to reduce market exposure temporarily.

In each case, a suitable derivative position may reduce some of the risk.

However, hedging involves costs and may also reduce profits if market prices move favourably.

Short hedge

A short hedge involves selling futures to protect the value of an asset or portfolio against a possible decline.

It is commonly considered when the participant owns an asset or expects to sell an asset in the future.

Example: An investor holding shares

Suppose an investor owns a diversified share portfolio worth ₹10,00,000.

The investor expects the market may decline over the next month but does not want to sell the shares because they are intended for long-term investment.

The investor sells suitable stock index futures.

Two possible outcomes may occur.

If the market falls: The share portfolio may lose value, while the short futures position may gain. The futures gain may partly offset the portfolio loss.

If the market rises: The portfolio may gain value, while the short futures position may lose.

The hedge is intended to reduce the effect of a market decline, not to guarantee a profit.

Long hedge

A long hedge involves buying futures to protect against an expected increase in the price of an asset that will be purchased in the future.

Example: A manufacturing company

A manufacturing company expects to purchase a commodity next month.

The current price is ₹1,000 per unit. The company is worried that the price may increase.

For illustration, assume it buys a suitable futures contract at ₹1,020 per unit.

If the futures price rises to ₹1,100, the long futures position gains ₹80 per unit.

The futures gain may offset some of the higher purchase cost.

If the futures price falls, the futures position may lose value, although the company may benefit from purchasing the commodity at a lower spot price.

The effectiveness of the hedge depends on how closely the futures price tracks the company's actual purchase price.

Difference between long and short hedging

Basis	**Long hedge**	**Short hedge**
Futures action	Buy futures	Sell futures
Main concern	Future purchase price may rise	Existing asset value may fall
Typical participant	Future buyer	Asset owner or future seller
Expected protection	Against price increases	Against price decreases
Example	Manufacturer expecting to buy a commodity	Investor holding shares

Hedging versus speculation

Basis	Hedging	Speculation
Purpose	Reduce an existing or expected risk	Seek a profit from market movements
Existing exposure	Usually present or anticipated	Not necessarily required
Risk effect	Intended to offset some other risk	Creates or increases market exposure
Example	Shareholder sells index futures to reduce downside risk	Trader buys index futures expecting a rise
Main objective	Risk management	Profit opportunity

Factors affecting the relationship between spot and futures prices

The difference between spot and futures prices can be affected by:

- Interest rates
- Storage and financing costs, where relevant
- Expected dividends for stock-related contracts
- Time remaining until expiry
- Supply-and-demand conditions
- Delivery requirements
- Transaction costs and market restrictions

For example, an index futures price may differ from the spot index because of financing costs and expected dividends during the remaining life of the contract.

MARK-TO-MARKET (MTM)

Meaning of mark-to-market

Mark-to-market, commonly called MTM, is the process of calculating gains or losses using the relevant current settlement price.

In exchange-traded futures, gains and losses are generally settled through the margin and clearing system according to applicable rules.

In simple language, the position is regularly valued at the current market price instead of waiting until the end of the contract to calculate its result.

Why is MTM important?

MTM is important because it:

1. Makes gains and losses visible as prices change.
2. Helps manage the risk of unpaid losses accumulating.
3. Supports the clearing and settlement process.
4. Helps participants monitor their positions.
5. Can require additional funds when the market moves against a position.

Functions of a clearing house

1. Clearing of trades

The clearing house determines the obligations arising from trades executed on the exchange.

2. Collection of margins

It collects or administers the required margins through the clearing framework to help manage potential losses.

3. Settlement of gains and losses

The clearing house manages the settlement of obligations arising from price movements and contract expiry.

4. Risk management

It monitors positions, exposures and margin requirements to manage the risk of participants failing to meet their obligations.

5. Default management

If a participant fails to meet its obligations, the clearing house follows established default-management procedures.

6. Improving confidence in the market

Common rules and settlement procedures help make exchange trading more organised and reliable.

How does a clearing house work?

The process can be understood through the following sequence.

1. A buyer and seller place orders through the exchange.
2. The exchange matches their orders.
3. The clearing corporation calculates the obligations arising from the trade.
4. Applicable margins are collected.
5. Gains and losses are calculated and settled according to the MTM process.
6. The final contract is settled at expiry according to its specifications.

STOCK INDEX FUTURES

A stock market index represents the performance of a selected group of shares according to a defined methodology.

Examples of well-known Indian stock market indices include:

- Nifty 50
- BSE Sensex
- Stock index futures are futures contracts based on a stock market index rather than a single company's share.
- They allow traders and investors to take positions on the broad market or hedge a portfolio against market movements.

Uses of stock index futures

1. Hedging an equity portfolio

An investor who owns shares may sell stock index futures to reduce exposure to a possible market decline.

2. Taking a market view

A trader who expects the index to rise may buy index futures.

A trader who expects the index to fall may sell index futures.

3. Temporary portfolio adjustment

A fund manager may use index futures to adjust market exposure while arranging changes to the underlying share portfolio.

4. Arbitrage

Traders may compare index futures with the theoretical value of the index futures contract and related share portfolio.